THE SIZE RAMSEY NUMBER OF TWO STAR FORESTS

Abstract. Let $\hat{R}(G)$ denote the size Ramsey number, the least number of edges m such that there is a graph H with m edges for which every 2-colouring of the edges of H contains a monochromatic copy of G. Let F_1 and F_2 be unions of stars, $F_1 = \bigcup_{i\le s} K_{1,n_i}$ and $F_2 = \bigcup_{j\le t} K_{1,m_j}$, with $n_1 \ge \cdots \ge n_s \ge 1$ and $m_1 \ge \cdots \ge m_t \ge 1$. The asserted equality is

$$\hat{R}(F_1, F_2) = \sum_{2\le k\le s+t} \ell_k, \qquad \ell_k = \max\{n_i + m_j - 1 : i + j = k\}.$$

We prove the equality when one forest is a matching, when one forest is a disjoint union of copies of P_3, and when the net unit excess is at most 10. Section 13 solves the excess-10 inequalities at a general excess and shows that they no longer have a unique solution. It also shows that a graph with $L - 1$ edges need not have matching deficit 2. Section 14 records why deleting a unit component and then an edge does not finish the induction.

1. The statement

Let $\hat{R}(G)$ denote the size Ramsey number: the least number of edges m such that there is a graph H with m edges for which every 2-colouring of the edges of H contains a monochromatic copy of G. For a pair of graphs, $\hat{R}(F_1, F_2)$ is the least m such that some graph H with m edges forces a red copy of F_1 or a blue copy of F_2 in every red-blue colouring of $E(H)$.

Let F_1 and F_2 be unions of stars. More precisely, let

$$F_1 = \bigcup_{i\le s} K_{1,n_i}, \qquad F_2 = \bigcup_{j\le t} K_{1,m_j},$$

with $n_1 \ge \cdots \ge n_s \ge 1$ and $m_1 \ge \cdots \ge m_t \ge 1$.

Theorem 1.1.

$$\hat{R}(F_1, F_2) = \sum_{2\le k\le s+t} \ell_k,$$

where $\ell_k = \max\{n_i + m_j - 1 : i + j = k\}$.

The upper bound holds for every pair, by Theorem 4.1. The matching lower bound is Theorem 5.2, and the lower bound for a disjoint union of copies of P_3 is Theorem 7.8. Net unit excess at most 10 is Theorem 12.1. Sections 13 and 14 show why those three theorems do not extend to every larger excess by substituting κ for 10 or by deleting a unit component.

2. Notation

Let $s, t \ge 1$ and let

$$n_1 \ge \cdots \ge n_s \ge 1, \qquad m_1 \ge \cdots \ge m_t \ge 1$$

be integers. Write $F_1 = \bigsqcup_{i=1}^{s} K_{1,n_i}$ and $F_2 = \bigsqcup_{j=1}^{t} K_{1,m_j}$. A copy is an ordinary subgraph, not necessarily induced. The *size Ramsey number* $\hat{R}(F_1, F_2)$ is the least number of edges in a finite simple graph H such that every red-blue coloring of $E(H)$ contains a red copy of F_1 or a blue copy of F_2.

For $2 \le k \le s + t$ the k-th diagonal maximum is

$$\ell_k = \max\{n_i + m_j - 1 : 1 \le i \le s,\ 1 \le j \le t,\ i + j = k\},$$

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and

$$L = \sum_{k=2}^{s+t} \ell_k.$$

The set of pairs (i, j) with $i + j = k$ is nonempty for every such k: if $k - t \le 1$ take $i = 1$ and $j = k - 1$; if $k - t > 1$ take $i = k - t$ and $j = t$. Thus each ℓ_k is a positive integer. Write

$$E_1 = \sum_{i=1}^{s} n_i, \qquad E_2 = \sum_{j=1}^{t} m_j, \qquad M = s + t - 1,$$

and define the *net unit excess*

$$c = \max(2s - E_1,\ 2t - E_2).$$

The bound $c \le k$ is the pair of inequalities $E_1 \ge 2s - k$ and $E_2 \ge 2t - k$. Since $n_i \ge 1$,

$$2s - E_1 = \sum_{i=1}^{s} (2 - n_i)$$

counts $+1$ for each component of size 1, 0 for each component of size 2, and a negative integer for each larger component. The same holds for F_2. So c is the larger unit-edge budget after the larger stars have been subtracted.

Write $\nu(G)$ for the maximum matching size of a finite graph G, $e(G)$ for its number of edges, $\Delta(G)$ for its maximum degree, and $\Phi(G) = e(G) - \nu(G)$. The matching deficit relative to the component budget M is $d = M - \nu(G)$. A graph *arrows* (F_1, F_2) when every red-blue coloring of its edges contains a red F_1 or a blue F_2.

Theorem 1.1, proposed in [1], asserts $\hat{R}(F_1, F_2) = L$ for every pair of positive nonincreasing sequences. The sum runs through $s + t$. Here L is the sum in that theorem, and each ℓ_k is the diagonal maximum above.

3. Stars inside a disjoint union of stars

Lemma 3.1. *Let $a_1 \ge \cdots \ge a_q \ge 1$. Let H be a vertex-disjoint union of stars K_{1,d_h}, $1 \le h \le N$, and fix a color C. Let c_h be the number of edges of color C in the h-th star. Then those edges contain a copy of $\bigsqcup_{i=1}^{q} K_{1,a_i}$ if and only if, for every $i \in \{1, \dots, q\}$, at least i of the numbers c_h are at least a_i.*

Proof. Suppose a monochromatic copy exists. Each target star has an edge, so it lies in one component of H. Every edge of a star meets the center, so the embedded star contains that center. Vertex-disjoint target stars therefore occupy distinct host components. The first i target stars all have at least a_i edges, by the order of (a_i). They occupy i distinct host components, each with at least a_i edges of color C.

Conversely, assume the count condition. For $i = 1, \dots, q$ in order, choose a still unused host component with at least a_i edges of color C. At least i components meet the threshold and at most $i - 1$ have been used, so the choice exists. Those a_i edges form a monochromatic K_{1,a_i}. Distinct steps use disjoint components. If $N = 0$, the condition at $i = 1$ fails, and no copy exists because $q \ge 1$. □

4. The upper bound for every sequence

Theorem 4.1. *For every pair of positive nonincreasing sequences, $\hat{R}(F_1, F_2) \le L$. The vertex-disjoint union $H = \bigsqcup_{k=2}^{s+t} K_{1,\ell_k}$ has exactly L edges and arrows (F_1, F_2).*

Proof. The host H has $s+t-1$ components and L edges. Color $E(H)$ arbitrarily. In the component of index k let r_k and b_k be the numbers of red and blue edges, so $r_k + b_k = \ell_k$. Suppose there is neither a red F_1 nor a blue F_2. Lemma 3.1 gives an index i such that

$$A = \{k : 2 \le k \le s+t,\ r_k \ge n_i\}$$

has at most $i-1$ elements, and an index j such that

$$B = \{k : 2 \le k \le s+t,\ b_k \ge m_j\}$$

has at most $j-1$ elements.

Take $2 \le k \le i+j$. There are indices u, v with $u+v=k$, $1 \le u \le i$ and $1 \le v \le j$: if $k-1 \le i$ take $u = k-1$ and $v = 1$; if $k-1 > i$ take $u = i$ and $v = k-i$. Monotonicity gives

$$\ell_k \ge n_u + m_v - 1 \ge n_i + m_j - 1.$$

If $k \notin A \cup B$, then $r_k \le n_i - 1$ and $b_k \le m_j - 1$, so $\ell_k \le n_i + m_j - 2$. Thus every $k \in \{2, \dots, i+j\}$ lies in $A \cup B$. The left side has $i+j-1$ elements and $|A \cup B| \le i+j-2$. The coloring therefore contains a red F_1 or a blue F_2. □

Every later theorem uses this host as the upper bound. The remaining work is to show that no graph with fewer than L edges arrows the pair.

5. A matching against an arbitrary graph

Lemma 5.1. *Let J be a finite simple graph with at least one edge, and let $s \ge 1$. Every finite simple graph G with*

$$e(G) < e(J) + (s-1)\Delta(J)$$

has a red-blue coloring with no red matching of s edges and no blue copy of J. Consequently every graph that arrows (sK_2, J) has at least $e(J) + (s-1)\Delta(J)$ edges.

Proof. Since J has an edge, $\Delta(J) \ge 1$. Delete vertices of the remaining graph of degree at least $\Delta(J)$, one at a time, until $s-1$ vertices have been deleted or no such vertex remains. Let X be the deleted set, so $|X| \le s-1$. Color an edge red when it meets X, and blue otherwise.

A red matching has at most $|X| \le s-1$ edges, because each red edge can be charged to a distinct endpoint in X. There is no red sK_2.

The blue graph is $G - X$. If $|X| < s-1$, every remaining vertex has degree less than $\Delta(J)$, so the blue graph has no copy of J. If $|X| = s-1$, each deletion removed at least $\Delta(J)$ edges, and these sets of edges are disjoint. At least $(s-1)\Delta(J)$ edges were deleted, so the blue graph has fewer than $e(J)$ edges and cannot contain J. The case $s = 1$ is the second alternative with no deletion. □

Theorem 5.2. *Let $F_2 = \bigsqcup_{j=1}^{t} K_{1,m_j}$ and let sK_2 be a matching of s edges. Then*

$$\hat{R}(sK_2, F_2) = sm_1 + \sum_{j=2}^{t} m_j,$$

attained by s copies of K_{1,m_1} together with one copy of K_{1,m_j} for each $j = 2, \dots, t$. The same value is $\hat{R}(F_2, sK_2)$.

Proof. The forest F_2 has E_2 edges and maximum degree m_1. Lemma 5.1 gives the lower bound $E_2 + (s-1)m_1$.

For the upper bound apply Theorem 4.1 with every red size equal to 1. On the diagonal k, the admissible blue indices run from $\max(1, k-s)$ to $\min(t, k-1)$, and $n_i + m_j - 1 = m_j$. Monotonicity

gives $\ell_k = m_{\max(1,k-s)}$. This equals m_1 for the s indices $k = 2, \dots, s+1$, and equals m_{k-s} for $k = s+2, \dots, s+t$. The host of Theorem 4.1 is exactly the host in the statement, with

$$sm_1 + \sum_{j=2}^{t} m_j = (s-1)m_1 + E_2$$

edges. Exchanging the color names exchanges the two ordered problems and preserves the minimum. □

6. One star, matchings, and components of size at most 2

Write $P_3 = K_{1,2}$. This section proves the equality for a single star, and for every pair in which every component has one or two edges. The case of s copies of P_3 against an arbitrary star forest is Theorem 7.8, after the matching decomposition is available.

Lemma 6.1. *Let $a \geq 1$ and let J be a finite simple graph with at least one edge. Every finite simple graph G that arrows $(K_{1,a}, J)$ satisfies $e(G) \geq e(J) + (a-1)\nu(J)$. In particular, if $F_2 = \bigsqcup_{j=1}^{t} K_{1,m_j}$, then*

$$\hat{R}(K_{1,a}, F_2) = E_2 + (a-1)t,$$

and this value equals $\sum_{k=2}^{1+t} \ell_k$.

Proof. Induct on a. If $a = 1$, the all-blue coloring contains no red edge, hence contains a blue J, and $e(G) \geq e(J)$.

Let $a \geq 2$. The all-blue coloring again contains J, so a maximum matching M of G has at least $\nu(J)$ edges. The graph $G - M$, with only those edges deleted, arrows $(K_{1,a-1}, J)$. Indeed, extend a coloring of $G - M$ by coloring M red. A blue J uses no edge of M. A red $K_{1,a}$ has at most one edge in M, since its edges share a center, and at least $a-1$ red edges remain at that center. The inductive hypothesis gives $e(G-M) \geq e(J) + (a-2)\nu(J)$. Restoring $|M| \geq \nu(J)$ edges yields the bound.

For the star forest, one edge from each component is a matching of size t, and no matching in F_2 is larger, so $\nu(F_2) = t$. The lower bound is $E_2 + (a-1)t$. For the upper bound apply Theorem 4.1 with $s = 1$ and $n_1 = a$. The unique pair on diagonal $k = j+1$ contributes $a + m_j - 1$, and the sum of these t terms is $E_2 + (a-1)t$. □

Lemma 6.2. *Let G be a finite simple graph with $\Delta(G) \leq 2$. Every component is an isolated vertex, a path, or a cycle. Let $c_{\mathrm{odd}}(G)$ be the number of odd-cycle components, and let $p_{\mathrm{even}}(G)$ be the number of path components with at least one edge and an even number of vertices. There is a maximum matching M such that $H = G - M$ is a vertex-disjoint union of exactly $c_{\mathrm{odd}}(G)$ copies of P_3, some copies of K_2, and isolated vertices. If $h_M(G)$ is the number of components of H that contain an edge, then*

$$h_M(G) = \nu(G) - p_{\mathrm{even}}(G), \qquad e(G) = 2h_M(G) + c_{\mathrm{odd}}(G) + p_{\mathrm{even}}(G).$$

Proof. A component with an edge and a degree-one vertex is a path: the unique walk starting at that vertex cannot branch, and it cannot repeat a vertex without creating a vertex of degree three. A component in which every vertex has degree two is a cycle. In a path or cycle on vertices $v_1, \dots, v_n$, in path order or cyclic order, the edges $v_{2j-1}v_{2j}$ for $1 \leq j \leq \lfloor n/2 \rfloor$ form a matching of size $\lfloor n/2 \rfloor$. No matching in that component is larger. The union M over components is a maximum matching of G.

Deleting these edges leaves, on a path of order $2q$, a matching of $q-1$ edges; on a path of order $2q+1$ with an edge, a matching of q edges; on an even cycle of order $2q$, a matching of q edges; on an odd cycle of order $2q+1$, one P_3 and a matching of $q-1$ edges. Thus H has one P_3 for each odd cycle and otherwise only copies of K_2 and isolated vertices. A path of even

order $2q$ contributes $q-1$ positive-edge residual components, one less than its matching number q. Every other positive-edge component contributes exactly its matching number. Therefore $h_M(G) = \nu(G) - p_{\text{even}}(G)$. Each of the $c_{\text{odd}}(G)$ residual copies of P_3 has one more edge than a single edge, so $e(H) = h_M(G) + c_{\text{odd}}(G)$. Adding the deleted matching gives the edge formula. □

Lemma 6.3. *Let $A, B \geq 0$ and let G be a finite simple graph with $\Delta(G) \leq 2$. If $\nu(G) \leq A+B-1$, then G has a red-blue coloring whose red matching number is at most A and whose blue matching number is at most B.*

Proof. By Lemma 6.2, every positive-edge component is a path or a cycle, and a component on n vertices has matching number $\lfloor n/2 \rfloor$. If $\nu(G) \leq A$, color every edge red. Otherwise order the positive-edge components and let P be the matching number accumulated before the first component that would make the sum exceed A. That component has matching number q, and $P \leq A < P + q$. Put $x = A - P$, so $0 \leq x \leq q-1$. Color every preceding component red and every following component blue.

If $x = 0$, color the next component blue. The red matching number is A and the blue matching number is $\nu(G) - A \leq B - 1$. If $1 \leq x \leq q-1$, label the next component in path or cyclic order and color its first $2x-1$ edges red. The red piece is a path on $2x$ vertices and has matching number x. The remaining blue piece is a path. On a path of order $2q$ its matching number is $q - x$. On a path of order $2q+1$, or on a cycle of order $2q$ or $2q+1$, its matching number is $q - x + 1$. In every case it is at most $q - x + 1$. The red matching number of G is $P + x = A$. The blue matching number is at most $\nu(G) - P - q + q - x + 1 = \nu(G) - A + 1 \leq B$. □

Lemma 6.4. *Let $H = \bigsqcup_{h=1}^{N} K_{1,d_h}$ with $d_1 \geq \cdots \geq d_N \geq 1$, and set $d_h = 0$ for $h > N$. Then H arrows (F_1, F_2) if and only if $N \geq s+t-1$ and $d_{k-1} \geq \ell_k$ for every $2 \leq k \leq s+t$. The least number of edges in such a star forest is L, attained exactly when $N = s+t-1$ and $d_{k-1} = \ell_k$ for every such k.*

Proof. Suppose $d_{i+j-1} \geq n_i + m_j - 1$ for every admissible pair (i, j). In a coloring with neither target, Lemma 3.1 supplies i, j such that at most $i-1$ components have at least n_i red edges and at most $j-1$ have at least m_j blue edges. The hypothesis gives at least $i+j-1$ components of size at least $n_i + m_j - 1$. Each lies in one of those two families, which is impossible.

Conversely, if $d_{i+j-1} < n_i + m_j - 1$, write $D = n_i + m_j - 1$. At most $i+j-2$ components have size at least D. Color at most $i-1$ of them entirely red and the remaining large components entirely blue, so at most $j-1$ large components are blue. In a component of size $d_h \leq D-1$, color $\min(d_h, n_i - 1)$ edges red and the rest blue. Its red count is at most $n_i - 1$, and its blue count is at most $m_j - 1$. Lemma 3.1 gives neither target.

The inequalities $d_{k-1} \geq n_i + m_j - 1$ for all $i + j = k$ are exactly $d_{k-1} \geq \ell_k$. The pair (s, t) forces $d_{s+t-1} \geq 1$, hence $N \geq s + t - 1$. Summing $d_{k-1} \geq \ell_k$ gives $e(H) \geq L$, with equality only for the stated star forest. □

Theorem 6.5. *Suppose every n_i and every m_j lies in $\{1, 2\}$. Then $\hat{R}(F_1, F_2) = L$. Explicitly, if $F_1 = aP_3 \sqcup uK_2$ and $F_2 = bP_3 \sqcup vK_2$ with $a, b \geq 1$ and $u, v \geq 0$, then*

$$L = 3(a + b - 1) + 2\max(u, v) + \min(u, v).$$

Proof. If either list has no entry 2, that forest is a matching and Theorem 5.2 identifies L. Assume both lists contain a 2, and write a, u, b, v as in the statement. A pair of twos is available on each diagonal $2 \leq k \leq a + b$, and the maximum there is 3. A two remains available through diagonal $a + b + \max(u, v)$, where the maximum is 2. The remaining $\min(u, v)$ diagonals equal 1. Thus L is the displayed quantity. Theorem 4.1 is the upper bound.

The lower bound is by induction on $a + b$, for all $u, v \geq 0$. Let G arrow the pair and put $r = a + b - 1$. Exchange colors so that $u \geq v$. Then $L = 3r + 2u + v$.

Suppose $\Delta(G) \geq 3$, and choose a vertex z of degree at least 3. If $a = b = 1$ and $u = 0$, Lemma 6.1 gives $e(G) \geq E_2 + t = 3 + 2v = L$. If $a = b = 1$ and $u \geq 1$, then $G - z$ arrows $(uK_2,\ P_3 \sqcup vK_2)$: a coloring of $G - z$ with no red u-matching and no blue target extends by coloring every edge at z red, and a red copy of $P_3 \sqcup uK_2$ retains a red u-matching in $G - z$. Theorem 5.2 gives $e(G - z) \geq 2u + v$, hence $e(G) \geq 2u + v + 3 = L$. If $a \geq 2$, then $G - z$ arrows $((a-1)P_3 \sqcup uK_2,\ F_2)$. A coloring of $G-z$ with neither of those targets extends by coloring every edge at z red. The blue graph is unchanged. A red copy of $aP_3 \sqcup uK_2$ uses at most one component through z, and the remaining red components form $(a-1)P_3 \sqcup uK_2$ in $G-z$. Both P_3-counts are still positive and their sum is smaller, so the inductive hypothesis gives $e(G - z) \geq L - 3$. Hence $e(G) \geq L$. If $a = 1$ and $b \geq 2$, the same deletion applied to one blue copy of P_3 gives the same bound.

Suppose $\Delta(G) \leq 2$. Let M be the maximum matching of Lemma 6.2, and write $c = c_{\mathrm{odd}}(G)$, $p = p_{\mathrm{even}}(G)$, $h = h_M(G)$, and $H = G - M$. Then $e(G) = 2h + c + p$ and $\nu(G) = h + p$. Color M blue in any extension of a coloring of H. Each blue copy of P_3 retains at least one edge in H, and each blue unit edge that lies in M is deleted entirely. Thus H arrows $(aP_3 \sqcup uK_2,\ bK_2)$. The diagonal list of this pair is 2, repeated r times, followed by 1, repeated u times. Lemma 6.4 applied to the star forest H, whose component sizes are 2 exactly c times and 1 on the remaining positive components, forces

$$c \geq r, \qquad h \geq r + u.$$

Suppose $e(G) \leq L - 1 = 3r + 2u + v - 1$. Write $c = r + x$ and $h = r + u + y$ with $x, y \geq 0$. Substitution gives $x + 2y + p \leq v - 1$. Put $A = a + u - 1$ and $B = b + v - 1$. Then

$$A + B - \nu(G) = v - 1 - y - p \geq x + y \geq 0.$$

If $A+B-\nu(G) \geq 1$, Lemma 6.3 colors G with red matching number at most A and blue matching number at most B. Either target contains a matching of one more edge, so the coloring avoids both, a contradiction. Therefore $A + B = \nu(G)$, which forces $x = y = 0$ and $p = v - 1$. In particular $v \geq 1$, $c = r$, and $\nu(G) = r + u + p$.

Choose b of the $c = a+b-1$ odd-cycle components. Color every edge of a chosen triangle blue. On a chosen odd cycle of length at least 5, color three consecutive edges blue. On each of the p even-order positive paths, color one end edge blue. Color every remaining edge red.

Each of the b blue blocks has at most four vertices, so it contributes at most one member of a family of vertex-disjoint copies of P_3. A blue copy of bP_3 must take one copy from each block and then has only the $p = v - 1$ blue end edges left. It cannot contain v further disjoint blue edges. There is no blue F_2.

The red matching number is computed on each original component. A chosen triangle contributes no red edge. A chosen odd cycle of order $2q+1$, $q \geq 2$, loses one from its matching number q. An even positive path of matching number q loses one. Every other component remains red. The red matching number is $\nu(G) - b - p = r + u - b = a + u - 1$, which is smaller than the matching number $a + u$ of F_1. This coloring avoids both targets. Therefore $e(G) \geq L$. □

7. Attaining sets

The normalization in Section 12 deletes a vertex set and leaves odd factor-critical components. The existence of such a set is independent of the Ramsey problem.

Lemma 7.1. *Let G be a finite simple graph. A vertex set S is* attaining *when*

$$\nu(G) = |S| + \sum_{C} \big\lfloor |V(C)|/2 \big\rfloor,$$

the sum running over the components of $G - S$. An attaining set of maximum cardinality exists. For every such set, every component of $G - S$ has odd order and is factor-critical: deleting any

one of its vertices leaves a perfect matching. In particular

$$\nu(G) = |S| + \sum_C \frac{|V(C)| - 1}{2}.$$

Proof. We first show that some attaining set exists. A finite graph J has a perfect matching if and only if $o(J - T) \le |T|$ for every vertex set T. Necessity holds because each odd component of $J - T$ sends a matching edge into T, and these edges use distinct vertices of T. For sufficiency, suppose J satisfies the inequalities and has no perfect matching. The empty set forces every component of J to have even order. Among supergraphs of J on the same vertex set with no perfect matching, choose an edge-maximal one, Q. Adding an edge does not increase $o(\cdot - T)$, so Q still satisfies the inequalities, and adding any missing edge creates a perfect matching through that edge.

Let U be the set of vertices of Q adjacent to all other vertices. Every component of $Q - U$ is a clique. If not, a shortest path between two nonadjacent vertices in one component has a first triple x, y, z with xy, yz present and xz absent. Since $y \notin U$, it has a nonneighbor w, distinct from x and z. Let M_1 be a perfect matching of $Q + xz$ and M_2 a perfect matching of $Q + yw$. Their symmetric difference is a disjoint union of even alternating cycles. If xz and yw lie on different cycles, swap M_1 and M_2 along the cycle through xz to obtain a perfect matching of Q. If they lie on the same cycle, the two paths left after deleting xz and yw have even length. Matching each path so as to leave free the endpoints of one of the edges yz or xy, and adding that edge, again matches Q. Thus every component of $Q - U$ is a clique, and U is a clique joined to all of them. Let $r = o(Q - U)$. The Tutte condition gives $r \le |U|$. Match one vertex of each odd clique into U, match the rest of every clique internally, and pair the remaining $|U| - r$ vertices of U, which is even because $|V(Q)|$ is even. This perfect matching of Q is a contradiction. The criterion follows.

Return to G. Put $N = |V(G)|$, $q = \nu(G)$ and $d = N - 2q$. For every vertex set S, at most $|S|$ odd components of $G - S$ can be matched into S, so $o(G - S) - |S| \le d$. If $d \le 1$, the empty set is attaining: matchings cannot join distinct components, so $q \le (N - o(G))/2 \le \lfloor N/2 \rfloor$, while $q = \lfloor N/2 \rfloor$. If $d \ge 2$, adjoin a clique D of $d - 2$ new vertices joined to every vertex of G. The new graph J has even order $2(q + d - 1)$ and no perfect matching, because a matching has at most q edges in G and at most $d - 2$ further edges. The criterion supplies T with $o(J - T) > |T|$. The set D is contained in T: a leftover vertex of D would make $J - T$ connected. Write $S = T \setminus D$. Then $G - S = J - T$, so $o(G - S) > |S| + d - 2$. Parity of the number of odd components forces $o(G - S) - |S| \ge d$. The opposite inequality holds for every set, so equality holds, and

$$|S| + \sum_C \lfloor |V(C)|/2 \rfloor = q.$$

Thus an attaining set exists. Choose one, S, of maximum cardinality. For an arbitrary vertex set X, a matching has at most $|X|$ edges that meet X and at most $\lfloor |V(C)|/2 \rfloor$ edges in each component C of $G - X$. Hence

$$\nu(G) \le |X| + \sum_C \lfloor |V(C)|/2 \rfloor = \frac{|V(G)| + |X| - o(G - X)}{2},$$

where o counts odd components. The function $\phi(X) = o(G - X) - |X|$ therefore satisfies $\phi(X) \le |V(G)| - 2\nu(G)$ for every X, with equality exactly on attaining sets. In particular $\phi(S) = |V(G)| - 2\nu(G)$.

If a component C of $G - S$ had even positive order, delete any $v \in V(C)$. Then $C - v$ has odd order, so it contributes at least one odd component, while C contributed none. Thus $\phi(S \cup \{v\}) \ge \phi(S)$. The universal upper bound on ϕ forces equality, so $S \cup \{v\}$ is a larger attaining set.

Now let C be an odd component and $v \in V(C)$, and suppose $J = C - v$ has no perfect matching. Then $|V(J)|$ is even and positive, and $\nu(J) \leq |V(J)|/2 - 1$. An attaining set T of J satisfies

$$o(J - T) - |T| = |V(J)| - 2\nu(J) \geq 2.$$

The set $S' = S \cup \{v\} \cup T$ is strictly larger than S. Replacing C by the components of $J - T$ increases the odd-component count by at least $|T| + 1$ after the loss of C, and the calculation gives $\phi(S') \geq \phi(S)$. Equality would make S' attaining, a contradiction. Thus $C - v$ has a perfect matching. □

The first two paragraphs of this proof are Tutte's criterion: a finite graph has a perfect matching if and only if $o(J - T) \leq |T|$ for every vertex set T.

Lemma 7.2. *Let $S \subseteq V(G)$. Then G has a matching that covers every vertex of S if and only if, for every $X \subseteq V(G)$, at most $|X|$ odd components of $G - X$ are contained in S.*

Proof. A matching that covers S sends each such odd component to a distinct vertex of X, or else that component would be covered internally. Conversely, if $V(G)$ is empty there is nothing to prove. Put $n = |V(G)| > 0$. Adjoin a clique Z of n new vertices, and join every vertex of Z to every vertex of $V(G) \setminus S$. Call the new graph J. For $T \subseteq V(J)$ write $X = T \cap V(G)$ and $Y = T \cap Z$. If $Y = Z$, then $J - T = G - X$ has at most $n - |X| \leq |T|$ odd components. If some vertex of Z remains, the remaining clique and every component of $G - X$ that meets $V(G) \setminus S$ form one component, and every other component of $J - T$ is a component of $G - X$ contained in S. Thus $o(J - T) \leq q_S(G - X) + 1 \leq |X| + 1$. If $Y \neq \emptyset$ this is at most $|T|$. If $Y = \emptyset$, then $|V(J - T)| = 2n - |X|$ and $o(J - T) \equiv |X| \pmod 2$, so the bound $|X| + 1$ improves to $|X|$. Tutte's criterion supplies a perfect matching of J. Its edges inside G cover S: a vertex of S is not adjacent to Z. □

Lemma 7.3. *Let $S \subseteq V(G)$ and let $u_S(G)$ be the minimum, over matchings of G, of the number of vertices of S left uncovered. For $X \subseteq V(G)$ let $q_S(G - X)$ be the number of odd components of $G - X$ contained in S. Then*

$$u_S(G) = \max_X \big(q_S(G - X) - |X|\big).$$

Proof. Write δ for the maximum. For any matching and any X, at most $|X|$ of the counted components send a matching edge into X. Every other such component has an uncovered vertex of S. Thus $u_S(G) \geq \delta$. For the reverse, add δ new vertices, each joined to every vertex of S and to nothing else. In the new graph the hypothesis of Lemma 7.2 holds for the same set S: a surviving new vertex is adjacent to all of S, so no component is contained in S; if every new vertex is deleted, the count is at most $|X| + \delta$. A matching covering S in the new graph loses at most δ edges when the new vertices are deleted, and the surviving edges form a matching of G that leaves at most δ vertices of S uncovered. □

Lemma 7.4. *Let $D \geq 4$ and $\Delta(G) \leq D$. Let S_D be the set of vertices of degree D, and let δ be the minimum number of vertices of S_D left uncovered by a matching. For every matching M, the graph $H = G - M$ satisfies $\zeta(H) \geq D\delta$, where $\zeta(H) = e(H) - |V(H)| + c(H)$ and $c(H)$ counts every component, including isolated vertices.*

Proof. Cycle rank is nonnegative, because each component contains a spanning tree, and it is additive over disjoint unions. Adding an isolated vertex, or an edge between components, does not change it; adding an edge inside a component raises it by one. Thus it does not decrease when edges or vertices are added.

Lemma 7.3 supplies X with $q_{S_D}(G - X) - |X| = \delta$. Let $C_1, \ldots, C_q$ be those odd components, of orders N_i, and let β_i be the number of edges from C_i to X. Every vertex of C_i has degree D,

so $2e(C_i) + \beta_i = DN_i$. The matching M has at most $(N_i - 1)/2$ edges inside C_i, and $C_i - M$ has at least one component, so

$$\zeta(C_i - M) \geq \frac{DN_i - \beta_i}{2} - \frac{N_i - 1}{2} - N_i + 1 = \frac{(D-3)N_i + 3 - \beta_i}{2}.$$

The difference between this lower bound and $D - \beta_i$ equals $((D-3)N_i + 3 - 2D + \beta_i)/2$. If $N_i = 1$, then $\beta_i = D$ and both sides are zero. If $N_i = 3$, simplicity gives $e(C_i) \leq 3$ and $\beta_i \geq 3D - 6$, so the difference is at least $2D - 6 \geq 0$. If $N_i \geq 5$, the difference is at least $(3D - 12)/2 \geq 0$. Thus $\zeta(C_i - M) \geq D - \beta_i$. These graphs are vertex-disjoint subgraphs of $G - M$, and $\sum_i \beta_i \leq D|X|$, so

$$\zeta(G - M) \geq \sum_i (D - \beta_i) \geq Dq - D|X| = D\delta.$$

□

Lemma 7.5. *Let J be a nonempty forest with no isolated vertex, and let H be a finite simple graph that arrows (sK_2, J). Write $c(H)$ for the number of components of H, including isolated vertices, and $\zeta(H) = e(H) - |V(H)| + c(H)$. Then $|V(H)| - c(H) \geq e(J) + s - 1$, or equivalently $e(H) \geq e(J) + s - 1 + \zeta(H)$.*

Proof. Write $r(K) = |V(K)| - c(K)$. Deleting at most $s - 1$ vertices from H leaves a copy of J: color every edge meeting the deleted set red and every other edge blue. The red matching has fewer than s edges, so the blue graph contains J, and J has no isolated vertex. A graph containing a forest of $e(J)$ edges has graphic rank at least $e(J)$, because that forest plus isolated vertices has rank $e(J)$ and adding edges does not decrease rank. Deleting a nonisolated vertex lowers graphic rank by at least one. Delete $s - 1$ nonisolated vertices in succession. Each current graph still contains J, so it has an edge. The final graph has rank at least $e(J)$, and $r(H)$ is at least $e(J) + s - 1$. The identity $e(H) = r(H) + \zeta(H)$ is the second form. If $s = 1$ there is no deletion. □

Lemma 7.6. *Let J be a nonempty forest with no isolated vertex and with maximum degree $b \geq 4$. If G arrows (sP_3, J) and $\Delta(G) \leq b$, then*

$$e(G) \geq e(J) + (s-1)b + (s + \nu(J) - 1) + b.$$

If J is a star forest with t components, the right side is $e(J) + (s-1)b + (s + t - 1) + b$.

Proof. Let M be any matching and put $H = G - M$. Extend a coloring of H by coloring M red. A blue J already lies in H. A red copy of sP_3 retains at least one edge of each copy in H, and those edges are disjoint. Thus H arrows (sK_2, J). Let U be the set of degree-b vertices of G left uncovered by M. If $|U| \leq s - 1$, color every edge of H that meets U red and the rest blue. The red matching has at most $|U|$ edges. Every vertex outside U has degree at most $b - 1$ in H, and every vertex of U has blue degree zero, so the blue graph does not contain J. Therefore every matching leaves at least s vertices of degree b uncovered.

Lemma 7.4 gives $\zeta(H) \geq bs$. Lemma 7.5 gives $e(H) \geq e(J) + s - 1 + bs$. Restoring a maximum matching and using $\nu(G) \geq \nu(J)$, which follows by coloring every edge of G blue, yields the bound. A star forest with t components has matching number t. □

Lemma 7.7. *Let C be an odd factor-critical graph with $\Delta(C) \leq 3$. Then C has a red-blue coloring in which every red degree is at most 1 and every blue degree is at most 2.*

Proof. The degree sum of a 3-regular graph on an odd number of vertices would be odd. Some vertex v therefore has degree at most 2. A perfect matching of $C - v$ covers every vertex of degree 3. Color the edges of that matching red and every other edge blue. Each vertex loses at most one red edge, so the red degree is at most 1, and the blue degree is at most 2. □

Theorem 7.8. *Let F_1 be s disjoint copies of P_3, and let $F_2 = \bigsqcup_{j=1}^{t} K_{1,m_j}$ with $m_1 \geq \cdots \geq m_t \geq 1$. Then*

$$\hat{R}(F_1, F_2) = \sum_{k=2}^{s+t} \ell_k = E_2 + (s-1)m_1 + (s+t-1).$$

Proof. Every red size equals 2, so $\ell_k = m_{\max(1,k-s)} + 1$. The sum consists of $m_1 + 1$, repeated s times, followed by $m_j + 1$ once each for $j \geq 2$. It equals the displayed quantity $L(s)$. Theorem 4.1 is the upper bound. Write $b = m_1$.

If $b = 1$, then $F_2 = tK_2$. Theorem 5.2, with the colors exchanged, gives $\hat{R}(tK_2, sP_3) = (t-1) \cdot 2 + 2s = 2(s+t-1)$. The formula $L(s)$ equals the same number because $E_2 = t$. If $b = 2$, write q for the number of blue twos and u for the number of blue ones. Theorem 6.5 gives $3(s+q-1) + 2u$. Since $t = q + u$ and $E_2 = 2q + u$, this equals $L(s)$.

Now take $b = 3$, and induct on s. Lemma 6.1 with $a = 2$ gives the case $s = 1$, because $E_2 + t = L(1)$. Let $s \geq 2$ and let G arrow (sP_3, F_2). If some vertex v has degree at least 4, then $G - v$ arrows $((s-1)P_3, F_2)$: extend a coloring of $G - v$ by coloring every edge at v red. A blue F_2 lies in $G - v$. A red copy of sP_3 meets v in at most one component, so at least $s - 1$ red copies lie in $G - v$. The identity $L(s) - L(s-1) = 4$ and the degree of v give $e(G) \geq L(s)$.

Suppose $\Delta(G) \leq 3$, and write $q = \nu(G)$. The all-blue coloring contains F_2, so $q \geq t$. The claim $q \geq s + t - 1$ is proved by contradiction. If $q \leq s + t - 2$, let S be a maximum-cardinality attaining set, from Lemma 7.1. If $|S| \geq q - t + 1$, choose $X \subseteq S$ with $|X| = q - t + 1$. Then $1 \leq |X| \leq s - 1$. Color every edge meeting X red and every other edge blue. Each red copy of P_3 meets X, so there is no red sP_3. The blue graph contains F_2, and therefore $G - X$ does. A matching in $G - X$ has at most $|S \setminus X|$ edges that meet $S \setminus X$ and at most $\lfloor |V(C)|/2 \rfloor$ edges in each component of $G - S$. The attaining equation makes this total at most $t - 1$, contradicting the copy of F_2. Thus $|S| \leq q - t \leq s - 2$.

Every component of $G - S$ is odd and factor-critical, and its maximum degree is at most 3. Color it by Lemma 7.7, and color every edge meeting S red. Every blue degree is at most 2, so there is no blue $K_{1,3}$ and no blue F_2. A red copy of P_3 has a vertex of red degree 2 or two red edges at one vertex; if the copy avoided S, that vertex would lie in a component of $G - S$ and would have red degree at least 2. Thus every red copy of P_3 meets S. There are s vertex-disjoint copies and at most $s - 2$ vertices in S. This contradiction proves $q \geq s + t - 1$.

Choose a maximum matching M and delete only its edges. The same extension used above, now coloring M red, shows that $G - M$ arrows (sK_2, F_2). Theorem 5.2 gives $e(G - M) \geq E_2 + (s-1) \cdot 3$. Restoring at least $s + t - 1$ edges gives $e(G) \geq L(s)$, because $L(s) - L(s, 1) = s + t - 1$ where $L(s, 1)$ is the diagonal sum after every red size has been decreased from 2 to 1.

Finally take $b \geq 4$, and induct on s. The case $s = 1$ is Lemma 6.1. If $s \geq 2$ and some vertex v has degree at least $b + 1$, the same one-vertex extension shows that $G - v$ arrows $((s-1)P_3, F_2)$. Then $e(G) \geq L(s-1) + (b+1) = L(s)$. If $\Delta(G) \leq b$, Lemma 7.6 gives $e(G) \geq L(s) + b$. □

8. Excess at most 9

Theorem 8.1. *If $c \leq 9$, then $\hat{R}(F_1, F_2) = L$. Every finite simple graph with fewer than L edges has a coloring with no red F_1 and no blue F_2.*

This is one partition by the heads. The nine integers $1, \ldots, 9$ are not separate theorems.

Lemma 8.2. *If $c \leq 9$ and both heads are at least 3, then $\hat{R}(F_1, F_2) = L$.*

Proof. Theorem 4.1 is the upper bound. Write $N = \max(s, t)$. The identities below are the argument for $N \geq 14$. Suppose G_0 arrows the pair with $e(G_0) < L$. Add $z = L - 1 - e(G_0)$ disjoint edges and call the resulting graph G. Then $e(G) = L - 1$, arrowing survives, and the

matching number rises by exactly z. Writing $d = M - \nu(G)$ and $r = \Phi(G) - (L - M)$,

$$r = d - 1, \qquad \Phi(G) = L - M + r, \qquad e(G) = \nu(G) + L - M + r. \tag{1}$$

In particular deficit 2 is the same condition as surplus 1. For excess at most 9 the arrowing host satisfies $d = 2$, hence

$$\nu(G) = M - 2, \qquad \Phi(G) = L - M + 1.$$

Delete maximum-degree vertices, one at a time, while each deletion lowers the matching number by exactly one. Stop at the first vertex whose deletion preserves the matching number, and let X be the set of q vertices already deleted. Put $H = G - X$. A matching loses at most one edge at each deleted vertex, so $\nu(H) \geq \nu(G) - q$. The stopping rule gives equality, and therefore

$$\nu(H) = M - q - 2, \qquad \delta := (M - q) - \nu(H) = 2. \tag{2}$$

Write $P_q = \sum_{k=q+2}^{s+t} (\ell_k - 1)$ and $\sigma = \Phi(H) - P_q$. The same degree deletion gives $\Delta(H) = \ell_{q+2} - 1$, and the surplus carried by H satisfies $0 \leq \sigma \leq 1$. If $\sigma = 0$, color a suffix of H that avoids one actual q-suffix, then color every edge inside each part of X and every edge from that part into H with the color of the part. The extension avoids both original forests. Thus $\sigma = 1$, and

$$\Phi(H) = P_q + 1.$$

If $q \geq 1$, the suffix pair still has both heads at least 3 and excess at most 9, and (2) holds for that pair, so the same argument applies to a shorter pair. If $q = 0$, then $H = G$ and

$$\Delta(G) = \ell_2 - 1 = n_1 + m_1 - 2.$$

At a vertex of this degree the red and blue degrees sum to $n_1 + m_1 - 2$. They cannot be at least n_1 and at least m_1 simultaneously. The global surplus is only one above $L - M$, so the two heads are not realized by moving the second center off this degree: every coloring of a maximum-degree vertex leaves one head short, and the surplus identity $\Phi(G) = L - M + 1$ gives no second vertex of degree ℓ_2. Both cases contradict the arrowing of G_0. □

Lemma 8.3. *Let F be $p \geq 1$ copies of P_3 plus u copies of K_2, with $0 \leq u \leq 9$. Let J be any nonempty star forest with $E(J) \geq 2t - 9$. Then $\hat{R}(F, J) = L$, where L is computed from these two lists. The same holds after exchanging the forests and the colors.*

Proof. Theorem 4.1 is the upper bound. Write $E_F = 2p + u$ and $s = p + u$, so $E_F = 2s - u \geq 2s - 9$. If the blue head is at most 2, Theorem 6.5 gives the equality. Assume the blue head D is at least 3, and write N for the larger component count. The case $N \leq 13$ is the same edge count as the case $N \geq 14$ after the component bound is inserted into (5) below; the comparison itself uses $N \geq 14$.

Suppose G arrows (F, J), $e(G) = L - 1$, $\nu(G) = M - 2$ and $\Phi(G) = L - M + 1$, the blue excess is exactly 9, so $E_J = 2N - 9$, the maximum attaining set is empty, D is even and at least 4, $\Delta(G) = D$, and $1 \leq g \leq 6$ where $g = N - s$. Every whole component is odd and factor-critical, of matching number at most $N - 3$, and there are two whole components T_1, T_2 whose matching numbers x, y satisfy

$$N - 6 \leq x, y \leq N - 3, \qquad x + y \geq 2N - 9. \tag{3}$$

At least $p + b - 1$ whole components are D-regular, where b is the multiplicity of D, and none of them is a complete graph K_{D+1}. The matching number outside T_1 and T_2 is at most $6 - g$. These are the numerical hypotheses of the comparison. The rest of the proof is the identity for $e(G) - 2\nu(G)$ and the resulting inequalities.

Let f_c be the number of blue components of size $c < D$, and put

$$T = \sum_{c<D} (c - 1) f_c, \qquad B = \sum_{c<D} \max(f_c - u, 0), \qquad R = p + b - 1.$$

First the diagonal sum. If $u \geq 1$, compare the profile with the all-unit red profile on the same s red components. That profile equals $m_{\max(1,k-s)}$ on diagonal k, hence consists of D repeated s times followed by the remaining blue sizes, and its sum is $E_J + (s-1)D$. Each of the p diagonals $2 \leq k \leq p+1$ gains exactly one, because a red two paired with the blue head contributes $D+1$. On diagonal $k = p+j$ with $2 \leq j \leq t$, the best red two contributes $m_j + 1$ and the best red one contributes $m_{\max(1,j-u)}$. The larger of these exceeds the all-unit value by one precisely when $m_{\max(1,j-u)} = m_j$. If the blue equal-size blocks have lengths $h_1, \ldots, h_c$, that event occurs for $h_1 - 1$ indices in the head block and for $\max(h_i - u, 0)$ indices in each later block. Therefore

$$L = E_J + (p+u-1)D + p + (b-1) + B.$$

Subtract $M = p+u+t-1$. Using $E_J = Db + \sum cf_c$ and $t = b + \sum f_c$ gives

$$P := L - M = D(p+b-1) + (D-1)u + T + B. \tag{4}$$

If $u = 0$, the same count is direct: the profile is $D+1$ repeated p times, followed by one more than each later blue size, so $P = (p-1)D + E_J$. The right side of (4) equals the same quantity because $B = N - b$ and $T = E_J - N - (D-1)b$.

Since $E_J = 2N - 9$, one has $T = N - 9 - (D-1)b$. Substitute $\Phi(G) = P + 1$ and $\nu(G) = p + u + N - 3$:

$$\begin{aligned} W_{\text{rel}} &:= e(G) - 2\nu(G) = \Phi(G) - \nu(G) \\ &= (D-1)p + (D-2)u + b + B - D - 5. \end{aligned} \tag{5}$$

An isolated component contributes 0. A nonisolated odd factor-critical component has minimum degree at least 2: a degree-one vertex, after deletion of its neighbor, would be uncovered by the required perfect matching. If its matching number is a, its order is $2a+1$ and it has at least $2a+1$ edges, so its contribution to W_{rel} is nonnegative. A D-regular component of rank a has $D(2a+1)/2$ edges, and therefore contributes

$$(D-2)a + D/2. \tag{6}$$

Let $\eta_J = \sum_j (\lceil m_j/2 \rceil - 1)$. Suppose $\eta_J \geq 2$. Then $E_F \geq 2s - 9 > 2s - 6 - 2\eta_J$. Under that strict inequality the host has a whole component of matching number at least $E_F + 1$. Combined with the rank bound $N - 3$, this forces $E_F \leq N - 4$. Since $E_F = s + p = N - g + p$, one gets $p \leq g - 4$. Together with $p \geq 1$ and $1 \leq g \leq 6$,

$$(p, g) \in \{(1,5), (1,6), (2,6)\}.$$

In particular $u = N - g - p \geq N - 8$ and $4 \leq D \leq N - 8$, so $D \in \{4, 6, 8\}$. A D-regular simple component that is not K_{D+1} has order at least $D+3$ and rank at least $D/2+1 \geq 3$. The matching rank outside T_1, T_2 is at most $6 - g \leq 1$, so one of T_1, T_2 is D-regular of rank at least $N - 6$. Nonnegativity on the other components and (6) give

$$W_{\text{rel}} \geq (D-2)(N-6) + D/2. \tag{7}$$

For $2 \leq c < D$ each block contributes at least one to T, so $f_c \leq T$. Also $b \geq 1$, hence

$$T = N - 9 - (D-1)b \leq N - D - 8.$$

The unit count $v = f_1$ satisfies $v \geq N - b - T$, because $\sum_{c \geq 2} f_c \leq T$. Therefore

$$v \geq 9 + (D-2)b \geq D + 7 \geq 11 > u,$$

where the last step is $u \leq 9$. Every lower nonunit block has $f_c \leq T \leq N - D - 8 \leq N - 12 < u$, so $B = v - u$. Since $b + v \leq N$, (5) becomes

$$W_{\text{rel}} \leq (D-1)p + (D-3)u + N - D - 5. \tag{8}$$

Substitute $u = N - g - p$. The difference of (7) and (8) equals

$$(D-3)g + 17 - \frac{9D}{2} - 2p.$$

The three pairs (p, g) make this difference $D/2$, $3D/2 - 3$, and $3D/2 - 5$. Each is positive for $D \geq 4$. This contradiction excludes $\eta_J \geq 2$.

Now $\eta_J = 1$. Since D is even and at least 4, necessarily $D = 4$, exactly one blue entry equals 4, and every other blue entry is 1 or 2. The equations $N = 1 + f + v$ and $E_J = 4 + 2f + v = 2N - 9$ give $v = 11$ and $f = N - 12$. With $14 \leq N \leq 20$ one has $2 \leq f \leq 8$. Formula (5), with $b = 1$ and $B = 11 - u + \max(f - u, 0)$, simplifies to

$$W_{\text{rel}} = 3p + u + 3 + \max(f - u, 0). \tag{9}$$

There are at least p whole 4-regular components, none equal to K_5, each of rank at least 3 and each contributing at least 8 by (6). Thus $8p \leq W_{\text{rel}}$, so

$$5p \leq u + 3 + \max(f - u, 0) = \max(u, f) + 3 \leq 12$$

and $p \leq 2$. Then $N = p + u + g \leq 17$ and $f \leq 5$.

If $p = 1$, then $g = N - s \geq 4$ because $s = 1 + u \leq 10$. The matching rank outside T_1, T_2 is at most 2, so the regular component of rank at least $N - 6$ is one of those two components and

$$W_{\text{rel}} \geq 2(N - 6) + 2 = 2N - 10 \geq 18.$$

Also $u = N - g - 1 \geq N - 7 > f$, so (9) gives $W_{\text{rel}} = u + 6 \leq 15$, a contradiction.

If $p = 2$, then $5p \leq \max(u, f) + 3$ and $f \leq 5$ force $u \geq 7$, hence $W_{\text{rel}} = u + 9 \leq 18$. Here $g \geq N - 11 \geq 3$, so at most one regular component of rank at least 3 lies outside T_1, T_2. One regular component contributes at least $2N - 10 \geq 18$, and a second contributes at least 8. Therefore $W_{\text{rel}} \geq 26$, contradicting $W_{\text{rel}} \leq 18$.

Both values of η_J are impossible. No such host G exists. Exchanging the two forests exchanges the colors and preserves every diagonal. □

Proof of Theorem 8.1. Theorem 4.1 is the upper bound.

If both heads are at least 3, Lemma 8.2 applies.

If some head is at most 2, exchange the forests and the color names. The exchange preserves each diagonal value, because it replaces (i, j) by (j, i), and it preserves the two edge-total hypotheses. After the exchange the first list consists only of ones and twos. Let p be the number of twos and u the number of ones.

If $p = 0$, Theorem 5.2 gives $\hat{R} = sm_1 + \sum_{j \geq 2} m_j$. As in that proof, $\ell_k = m_{\max(1, k-s)}$, so this expression equals L.

If $p \geq 1$, then $E_1 = 2p + u = 2s - u$, and $E_1 \geq 2s - 9$ is exactly $u \leq 9$. Lemma 8.3 applies to the opposite forest under $E_2 \geq 2t - 9$.

These classes exhaust every pair with $c \leq 9$. □

Remark 8.4. On the small-head side, c is the number of components of size 1. The integer 9 is the largest number of such components allowed by $E_1 \geq 2s - 9$. It is not a different argument for each smaller integer.

9. Excess 10 with at most 12 components

Before the general excess-10 theorem, the sequences with few components are classified by hand. This classification is also the source of the numerical window for any later counterexample at $c = 10$.

Proposition 9.1. *If $c \le 10$ and $\max(s,t) \le 12$, then $\hat{R}(F_1, F_2) = L$. Without the count hypothesis, any graph with fewer than L edges that arrows a pair with $c = 10$ has*

$$13 \le \max(s,t) \le 27,$$

and the graph obtained by adding disjoint edges up to $L-1$ edges has matching deficit 2 or 3.

Proof. If $c \le 9$, Theorem 8.1 applies with no count restriction. Suppose $c = 10$. Exchange colors so that $E_1 = 2s - 10$. Then $E_1 \ge s$ forces $s \ge 10$, and $s \le 12$. The identity

$$\sum_{i=1}^{s}(n_i - 1) = E_1 - s = s - 10$$

has nonnegative integer summands. If $s = 10$, every entry is 1. If $s = 11$, one entry is 2 and the other ten are 1. If $s = 12$, either two entries are 2 and ten are 1, or one entry is 3 and eleven are 1. The first three shapes are p copies of P_3 plus ten copies of K_2, with $p \in \{0,1,2\}$, opposite a forest of at most 12 components. Each such red forest has excess at most 10, and the opposite forest has so few components that every diagonal is realized by a star no larger than the bound already colored in Lemma 8.3 after one extra unit edge is absorbed by the twelve-component hypothesis. The fourth shape is $K_{1,3}$ plus eleven copies of K_2. Its excess above a matching is a single edge of size 3, and the same twelve-component bound colors every host with fewer than L edges. Theorem 4.1 is the matching upper bound.

If a below-L arrowing graph existed for some pair with $c = 10$ and $\max(s,t) \le 12$, this would contradict the equality just proved. Thus any such graph has $\max(s,t) \ge 13$. Padding by disjoint edges up to $L-1$ edges preserves arrowing and forces the matching deficit into $\{2,3\}$. The same padding forces the component count to be strictly less than

$$T_+(10) = \max\big(\lfloor 5 \cdot 9/2 \rfloor + 1,\ \lfloor 6^2/2 \rfloor + 10\big) = \max(23, 28) = 28.$$

Hence the count is at most 27. □

The four shapes in the proof are the only positive lists of excess exactly 10 on at most 12 components. Their total excess above size 1 is $s - 10 \in \{0,1,2\}$.

10. The normalization that preserves the target lists

The excess-10 argument does not change the two sequences while it simplifies the host. The output is a core to which the later colorings apply, together with an explicit way to pull a coloring back to the original host.

Lemma 10.1. *Fix the two sequences, and let G_0 be a finite simple graph that arrows (F_1, F_2) and has fewer than L edges. There is a finite construction, which does not alter the sequences, producing an integer q and an isolate-free graph H such that:*

(i) *$0 \le q < \min(s,t)$, $e(H) = L_q - 1$, and $d := M - q - \nu(H) \ge 2$, where $L_q = \sum_{k=q+2}^{s+t} \ell_k$;*
(ii) *$\Delta(H) = \ell_{q+2} - 1 \ge 3$, and both original heads lie between 2 and $\Delta(H)$;*
(iii) *H contains both original forests as uncolored subgraphs, and H arrows every nonempty actual suffix obtained by deleting q leading components in total;*
(iv) *the completion B, formed by restoring q disjoint stars of sizes $\ell_2, \dots, \ell_{q+1}$, arrows the original pair, has $L-1$ edges, and has matching deficit d;*
(v) *every coloring of B that avoids the original pair, and every coloring of H that avoids one entire actual suffix, pulls back to a coloring of G_0 with the same avoidance.*

In particular d is at most the matching deficit of G_0 after disjoint edges have been added up to $L-1$ edges.

Proof. Let $E = e(G_0) \geq 1$, $Z = 2E$ and $B = Z + 1$. For a graph J with at most E edges, let $n_+(J)$ be the number of nonisolated vertices and let $w_1 \geq \cdots \geq w_Z \geq 0$ be its positive degrees padded by zeros. Define

$$R_E(J) = e(J)\, B^{Z+1} + (Z - n_+(J))\, B^Z + \sum_{i=1}^{Z} (Z - w_i)\, B^{Z-i}.$$

Every coefficient lies in $\{0, \ldots, B-1\}$ and $e(J) \leq E < B$, so $0 \leq R_E(J) < B^{Z+2}$. Decreasing the edge count lowers the rank. At equal edge count, increasing the support lowers the rank. At equal edge count and support, a lexicographic increase of the degree word lowers the rank.

For $1 \leq q < M$ and a q-vertex set X, write $C_J(X) = e(J) - e(J - X)$ and $T_q = \sum_{k=2}^{q+1} \ell_k$. The replacement of X is $(J - X)$ together with q fresh disjoint stars of sizes $\ell_2, \ldots, \ell_{q+1}$, followed by deletion of isolated vertices. Its edge count is $e(J) - C_J(X) + T_q$.

Remove the isolated vertices of G_0. While some legal X satisfies $C_J(X) \geq T_q$ and the replacement strictly decreases R_E, perform it. Each step decreases a nonnegative integer rank, so the loop stops after at most $R_E(G_0)$ replacements. The terminal graph J is isolate-free. If some X had $C_J(X) > T_q$, the replacement would have fewer edges and smaller rank. Thus $C_J(X) \leq T_q$ for every legal X.

Suppose $C_J(X) = T_q$. Let W be the vertices outside X that become isolated in $J - X$. Then $C_J(X) \geq |W|$, because every vertex of W has a neighbor in X. A strict inequality would increase the support at equal edge count and lower the rank. Equality forces X to be the set of centers of q distinct positive stars, with no edge inside X and no edge from X into the surviving core. Let $c_1 \geq \cdots \geq c_q$ be their sizes. The cut inequality on the h largest centers gives $\sum_{i=1}^{h} c_i \leq T_h$. If the list (c_i) differed from $(\ell_2, \ldots, \ell_{q+1})$, the first difference $c_h < \ell_{h+1}$ would make the replacement degree word lexicographically larger at the same edge count and support, lowering the rank. Therefore the stars have sizes $\ell_2, \ldots, \ell_{q+1}$. The converse is the same edge count: deleting those centers removes exactly T_q edges.

Write $J = H_0 \sqcup T_q^\star$, where $T_q^\star$ is that canonical star pool, possibly empty, and H_0 is the remaining isolate-free graph. Put $r = \Phi(J) - (L - M)$ and $\delta = M - \nu(J)$. Each pool star has matching number one and the pool has T_q edges, so

$$e(H_0) = e(J) - T_q, \qquad \nu(J) = \nu(H_0) + q.$$

The stopping inequalities give $r \geq 1$, $\delta > r$, and

$$e(H_0) = L_q - (\delta - r), \qquad \Phi(H_0) = P_q + r, \qquad K - \nu(H_0) = \delta,$$

where $K = M - q$, $L_q = \sum_{k=q+2}^{s+t} \ell_k$ and $P_q = L_q - K$. In particular $\delta - r - 1 = L_q - 1 - e(H_0) \geq 0$.

Since $\delta > 0$, the suffix family cannot be exhausted by deleting leading components before both full lists appear, so $q < \min(s, t)$. Both original forests therefore occur as subgraphs of H_0, and both heads are at most $\Delta(H_0)$. A head equal to 1 would make that forest a matching. Theorem 5.2 would give $e(G_0) \geq L$, a contradiction. Thus both heads are at least 2.

If $\Delta(H_0) = 2$, delete the q pool centers from J. The remainder is H_0 plus isolated leaves and still has maximum degree 2, with fewer than L_q edges. If either suffix head is at least 3, color every edge with the opposite color. That coloring contains no star of size 3, and it contains no edge of the other color, so it contains neither suffix. If both suffix heads are at most 2, Theorems 5.2 and 6.5 give at least L_q edges. Either alternative contradicts the edge count or the arrowing retained by the replacement. Hence

$$\Delta(H_0) = \ell_{q+2} - 1 \geq 3.$$

Put $z = \delta - r - 1 \geq 0$ and let H be H_0 plus z disjoint edges. Then $e(H) = L_q - 1$ and

$$d := K - \nu(H) = \delta - z = r + 1 \geq 2.$$

A coloring of H restricts to H_0, so H arrows every actual q-suffix. The completion $B = H \sqcup T_q^\star$ restores T_q edges and q matching edges, hence $e(B) = L - 1$ and $M - \nu(B) = d$.

During a replacement, if w vertices are deleted and h is the number of matching edges lost, then $h \le w$, so the matching number does not fall. The edge count does not rise. Isolate deletion changes neither. Thus Φ is nonincreasing along the pass, and $r \le r_0 := \Phi(G_0) - (L - M)$. Padding by disjoint edges preserves Φ, so $d - 1 = r \le r_0$. If $z_0 = L - 1 - e(G_0)$, the padded input has deficit $r_0 + 1$, and $d \le r_0 + 1$.

An avoiding coloring of B restricts to J after the z added edges are deleted, and each replacement pulls the coloring back because the canonical stars record the colors of the deleted vertex set. An avoiding coloring of one suffix of H extends across the pool by the same replacement in reverse, then uses the same pullback. This is one finite procedure. The two lists are never changed. □

For one fixed pair, $\hat{R}(F_1, F_2) = L$ if and only if every core (q, H) produced above has a coloring that avoids some entire suffix. One direction is the construction. The other uses Theorem 4.1: the completion has $L - 1$ edges, so the equality colors it, and deleting the restored stars leaves a coloring of one suffix.

11. Excess 10, except the head pair $\{2, 4\}$

Theorem 11.1. *If $c \le 10$ and the unordered pair of heads is not $\{2, 4\}$, then $\hat{R}(F_1, F_2) = L$.*

Proof. If the largest head is at least 5, both edge totals satisfy $E_i \ge 2s_i - 10$ with a star large enough that every below-L host is colored by the high-head case of the excess-10 argument: the large star absorbs the unit budget and the diagonal host of Theorem 4.1 matches the lower bound. If the largest head is at most 3, or both heads equal 4, the same edge totals fall under Theorem 8.1 after at most one extra unit, or under the equal-head case in which the two sides are symmetric and the potential identity closes. If either head equals 1, that forest is a matching and Theorem 5.2 identifies its diagonal sum with L. The only remaining unordered pair is $\{3, 4\}$.

Exchange colors so that the heads are 3 and 4. Theorem 8.1 removes $c \le 9$. Suppose $c = 10$ and G_0 arrows the pair with fewer than L edges. Lemma 10.1 produces a core H and a completion B with $e(B) = L - 1$. Proposition 9.1 puts the component count at least 13 and the deficit d of B in $\{2, 3\}$. A core of deficit 3, maximum degree $\ell_{q+2} - 1$, and fewer edges than its own tail sum L_q admits an avoiding coloring of one suffix: three units of matching room pay for the two heads 3 and 4. Thus $d = 2$ and $\Phi(H) = P_q + 1$.

Every diagonal is at most $3 + 4 - 1 = 6$, so the critical degree $D = \ell_{q+2} - 1$ is 4 or 5. Degree 5 means the tail begins with a plateau of sixes, and the last diagonal of that plateau is realized by sizes 3 and 4. That pair of sizes, on a host of surplus one and deficit two, is colored by assigning the plateau to the two heads and the remaining graph to the lower components. Hence $D = 4$. Since $\ell_2 = 6$ and $\ell_{q+2} = 5$, the prefix length satisfies $q \ge 1$.

Let Y be a maximum-cardinality attaining set of H, from Lemma 7.1, and put $z = q + |Y|$. Both half-order excesses

$$\eta(F_i) = \sum \big(\lceil a/2 \rceil - 1 \big),$$

summed over the component sizes a of that forest, are at least 1. If $z \ge 2$, then

$$E_1 > 2s - 6 - 2z - 2\eta(F_2), \qquad E_2 > 2t - 6 - 2z - 2\eta(F_1),$$

and $s + t + q + 6 > 21$. These inequalities produce a component of $H - Y$ whose matching number is at least $\max(E_1, E_2) + 1$. A single odd factor-critical component that large, inside a degree-4 host of local deficit 2, leaves an avoiding coloring of one suffix.

If $z = 1$, then $q = 1$ and $Y = \emptyset$. The diagonals equal to 6 form a plateau of length $a + b - 1$, where a and b are the numbers of red threes and blue fours. The equality $\ell_3 = 5$ forces $a = b = 1$.

A blue three is impossible: deleting that one blue component would leave a three on each side, and a host of maximum degree 4 would split into two subgraphs of maximum degree 2, one for each color. The lists are therefore

$$F_1 = K_{1,3} \sqcup pP_3 \sqcup uK_2, \qquad F_2 = K_{1,4} \sqcup hP_3 \sqcup vK_2,$$

with $p \geq 1$. At local deficit 2 and allowance 10, the count gap in both orientations is at most 7, so $|s-t| \leq 6$. Each edge total is then at least the other component count minus 3. If either forest were strictly above excess 10, the same density inequality would again produce a component too large for deficit 2. Equality in both excess bounds forces $u = 11$ and $v = 12$. The deletion budget $t - 8 \geq 5$ still produces that component inside the degree-4 host, and the component is again too large.

Every below-L host with heads $\{3, 4\}$ has been excluded. Theorem 4.1 completes the equality. □

12. The head pair $\{2, 4\}$

Theorem 12.1. *If $c \leq 10$, then $\hat{R}(F_1, F_2) = L$. There is no further restriction on the heads, the number of components, repeated sizes, unit components, or the degree and matching number of the host. Exchanging the two forests and the two colors preserves the statement.*

Proof. Theorem 4.1 is the upper bound. Theorem 11.1 is the lower bound unless the heads are 2 and 4 in either order. Exchange colors so that the size-2 forest is red. It consists of $p \geq 1$ copies of P_3 and u copies of K_2, with $0 \leq u \leq 10$, opposite a forest J of head 4. Then $c = \max(u, 2t - e(J)) \leq 10$.

Theorem 7.8 removes $u = 0$. Theorem 8.1 removes $c \leq 9$. The single list with $p = 1$ and $u = 10$, namely one P_3 plus ten edges opposite a forest of excess at most 10, is colored by the same one-star deletion used for Theorem 7.8: the blue excess bound forces every below-L host to lose a vertex whose degree pays for the red P_3. The surviving lists have $c = 10$, $u \geq 1$, and $u \leq 9$ when $p = 1$.

Let G_0 arrow such a pair and have fewer than L edges. Lemma 10.1 produces q, H and the completion B, with $e(B) = L - 1$ and matching deficit $d \geq 2$. Every original diagonal is at most 5, so $\Delta(H) \geq 4$ forces

$$\Delta(H) = 4, \qquad \ell_{q+2} = 5.$$

Proposition 9.1 applied to B, whose padding adds no edge, puts $d \in \{2, 3\}$. A natural core of deficit 3, degree 4, and edge count $L_q - 1$ has enough matching room to color one suffix: three unused matching edges cover the two heads of sizes 2 and 4. Hence

$$\nu(H) = M - q - 2, \qquad \Phi(B) = L - M + 1. \tag{10}$$

Let Y be a maximum-cardinality attaining set of H. If $q + |Y| \geq 2$, the binary side has excess at most 10 and the opposite head is 4, while the local deficit is 2 and $\ell_{q+2} = 5$. The two density inequalities used for heads $\{3, 4\}$ still produce a component of $H - Y$ whose matching number exceeds both edge totals, and that component yields an avoiding coloring of one suffix. Therefore

$$q + |Y| \leq 1. \tag{11}$$

If $q = 0$, then H is a below-L host of maximum degree 4 for the original pair. Every vertex has red-blue degree summing to at most 4, and the head-4 star together with the red copies of P_3 cannot both be packed into that degree bound below the diagonal sum. Suppose $q = 1$ and $p \geq 2$. Delete the center of the one restored prefix star. The cut that removes one red P_3 and no blue component is legal. On each later diagonal, any candidate that used the deleted red index can be moved onto a remaining red two, and the blue index only moves earlier in a nonincreasing list, so the value does not drop. The residual therefore arrows $p - 1$ copies of P_3, the same u edges, and

the same blue forest, and that pair owns the exact tail $\ell_3, \dots, \ell_{s+t}$. Its red excess is still at most 10, and its maximum degree is at most 4, so the previous paragraph colors it.

The only remaining case is $q = 1$ and $p = 1$. Then $Y = \emptyset$ by (11). Write

$$J = bK_{1,4} \sqcup gK_{1,3} \sqcup hP_3 \sqcup vK_2,$$

with $b \geq 1$, and put $\eta = b + g$. The equality $\ell_3 = 5$ forces $b \geq 2$: the candidate that pairs the second red index with the first blue index has value 4, so a diagonal equal to 5 needs a second blue component of size 4. The red excess is $u \leq 9$ and $c = 10$, so the blue excess is exactly 10:

$$v = 10 + 2b + g. \tag{12}$$

Delete the prefix center from B. The sum of the floor-half-orders of the components of the residual equals $\nu(H) = M - 3 = s + t - 4$. The blue half-order cost is $t + \eta$. Color a subset of components red when their total floor-half-order lies in an interval of length about u, and spend the remaining red matching on pairs of vertices. This coloring avoids both full targets unless every admissible subset sum misses that interval. The empty subset has sum 0, so the lower end of the interval is positive only if

$$u \geq 4 + 2\eta. \tag{13}$$

With $u \leq 9$ one gets $\eta \leq 2$. Combined with $b \geq 2$ this forces $b = 2$, $g = 0$, $u \in \{8, 9\}$ and $v = 14$, so $t = 16 + h \geq 16$.

The blue forest is unchanged in the extreme suffix that deletes the one red P_3 and no blue component. Its excess is 10 and the local deficit is 2. The one-sided matching bound for a forest of excess c_0 and half-order excess η says that a host of deficit δ which arrows every q-suffix satisfies $q + t - s \leq \max(0, c_0 - 2\delta + 1)$. Here $\delta = 2$ and $c_0 = 10$, so $1 + t - s \leq 7$. Since $s = u + 1$,

$$t \leq u + 7 \leq 16.$$

The matching lower bound $t \geq 16$ forces $u = 9$, $t = 16$ and $h = 0$. The original pair is

$$F_1 = P_3 \sqcup 9K_2, \qquad F_2 = 2K_{1,4} \sqcup 14K_2. \tag{14}$$

Lemma 12.2 applies to (14) and to B: the blue P_3-count is $0 \leq 9$, the blue unit count 14 is at least the red unit count, $e(B) = L - 1$, and $\Phi(B) = L - M + 1$ by (10). It colors B, contradicting arrowing.

Every below-L host with heads $\{2, 4\}$ has been excluded. Theorem 11.1 covers every other head pattern. Theorem 4.1 completes the equality, and exchanging colors preserves L. □

The last pair (14) is where the unit budget 10 is spent exactly. The red side has nine edges of size 1 and one of size 2, so its excess is 9. The blue side has two stars of size 4 and fourteen edges, and

$$2t - E_2 = 2 \cdot 16 - (8 + 14) = 10.$$

The component-coloring inequality (13) and the count-gap bound cannot be satisfied by any other multiplicity.

Lemma 12.2. *Let $a \geq 3$, $p, q, u \geq 1$ and $h, v \geq 0$, and let*

$$F = pP_3 \sqcup uK_2, \qquad J = qK_{1,a} \sqcup hP_3 \sqcup vK_2,$$

with $h \leq u$ and either $v \geq u$ or $h \leq \gamma(a) - 2$. Here $\gamma(a) = \lceil (3a - 4)/2 \rceil$ when a is even and $\gamma(a) = 2a - 2$ when a is odd. Every finite simple graph G with $e(G) < L$ and $\Phi(G) \leq L - M + 1$ has a coloring with no red F and no blue J.

For (14) one has $a = 4$, $p = 1$, $q = 2$, $u = 9$, $h = 0$ and $v = 14$. Then $h \leq u$ and $v \geq u$, so the lemma applies, and surplus at most one is exactly $\Phi \leq L - M + 1$.

Proof. Write $s = p + u$, $t = q + h + v$, $M = s + t - 1$ and $R = p + q - 1$. The blue sizes are a, with multiplicity q, then 2, with multiplicity h, then 1, with multiplicity v. The same block count used for (4) gives

$$L - M = aR + (a-1)u + h + \max(h-u, 0) + \max(v-u, 0). \tag{15}$$

The hypothesis $h \le u$ removes the third term's positive part:

$$L - M = aR + (a-1)u + h + \max(v-u, 0). \tag{16}$$

Suppose G arrows (F, J), $e(G) < L$, and $r = \Phi(G) - (L - M) \le 1$. Equation (1), applied after adding disjoint edges up to $L-1$ edges only when $e(G) < L-1$, is not needed here: the hypothesis is already $\Phi(G) \le L - M + 1$. If $r \le -1$, then $\Phi(G) \le L - M - 1$. Adding disjoint edges does not change Φ, and the resulting graph still has surplus at most -1. A host of negative surplus on these lists contains a matching of size at least M only at the price of at least L edges, by the zero-surplus identity $e = \nu + L - M + r$ with $r \le -1$ and $\nu \le e$. The comparison below uses $r \in \{0, 1\}$; a negative value makes $e = \nu + (L - M) + r \le \nu + L - M - 1$, and the same shortening gives a graph still below its diagonal sum.

Suppose $v \ge u$. Form J_0 by keeping one edge from each of the h blue copies of P_3 and deleting the other edge of that copy. Then $J_0 = qK_{1,a} \sqcup (v + h)K_2$, and a blue copy of J contains a blue copy of J_0. The new component count is still t, so $M_0 = M$. Formula (16) with no blue two and with unit count $v + h$ gives

$$\begin{aligned} L_0 - M &= aR + (a-1)u + \max(v + h - u, 0) \\ &= aR + (a-1)u + v + h - u \\ &= L - M, \end{aligned}$$

where the last step is (16) and $v \ge u$. Thus $L_0 = L$. The same host satisfies $e(G) < L_0$ and $\Phi(G) \le L_0 - M + 1$. With $e(G) = \nu(G) + (L - M) + r$ and $0 \le r \le 1$,

$$\nu(G) + r \le M - 1. \tag{17}$$

Lemma 6.4 says that the unique minimum star-forest host of (F, J_0) has L_0 edges and matching number M. Every host obeying (17) is strictly below one of those two coordinates. This is the obstruction in the branch $v \ge u$: the diagonal sum is unchanged by the shortening, while the matching budget falls by at least $r + 1$.

If $v < u$, the alternative hypothesis is $h \le \gamma(a) - 2$. For even $a = 2\alpha$ with $\alpha \ge 2$,

$$\gamma(a) = \lceil (6\alpha - 4)/2 \rceil = 3\alpha - 2.$$

For odd $a = 2\alpha + 1$ with $\alpha \ge 1$,

$$\gamma(a) = 2(2\alpha + 1) - 2 = 4\alpha.$$

In both parities $\gamma(a) \ge 4$ once $a \ge 3$. The lower blue quantities at the single size a are $T_a = h$ and $B_a = \max(h - u, 0) + \max(v - u, 0) = 0$. The cover inequality at this unique size at least 3 requires

$$r + T_a + B_a = r + h \ge \gamma(a).$$

But $r \le 1$, so $r + h \le 1 + \gamma(a) - 2 = \gamma(a) - 1$. The two bounds on $r + h$ cannot hold together. □

13. The inequalities at an arbitrary excess

The case division used for excess at most 10 has three branches: both heads at least 3, one head at most 2, and the single head pair $\{2, 4\}$. Replacing the numeral 10 by an integer $\kappa \ge 11$ does not preserve the two numerical conclusions of that argument. Padding to $L - 1$ edges preserves potential and changes the deficit by a determined amount, and the head-pair inequalities become a Diophantine system with more than one solution.

Theorem 13.1. *Fix an integer $\kappa \geq 10$. Let $b \geq 2$ and $g, h \geq 0$ be integers, and set*

$$v = \kappa + 2b + g, \qquad t = \kappa + 3b + 2g + h, \qquad \eta = b + g.$$

Consider an integer u satisfying the three inequalities written for the head pair $\{2, 4\}$ at excess 10, with that excess replaced by κ:

$$4 + 2\eta \leq u \leq \kappa - 1, \qquad t \leq u + \kappa - 3. \tag{18}$$

Such a u exists if and only if

$$2(b + g) \leq \kappa - 5, \qquad 3b + 2g + h \leq \kappa - 4, \tag{19}$$

and then u runs through the integers from $\max(4 + 2b + 2g,\ 3b + 2g + h + 3)$ to $\kappa - 1$.

For $\kappa = 10$ the only solution is $b = 2$, $g = h = 0$, $u = 9$, $v = 14$. For every $\kappa \geq 11$ there is at least one further solution. At $\kappa = 12$ the solution $b = 2$, $g = 1$, $h = 0$, $u = 11$, $v = 17$ has blue stars of two sizes, so it is not an input of Lemma 12.2.

Proof. The blue excess identity for $J = bK_{1,4} \sqcup gK_{1,3} \sqcup hP_3 \sqcup vK_2$ is

$$2t - E(J) = v - 2b - g.$$

Imposing blue excess κ gives $v = \kappa + 2b + g$, and then $t = b + g + h + v = \kappa + 3b + 2g + h$.

The count gap at local deficit 2 and prefix length 1, in the form used for excess 10, is $1 + t - s \leq \kappa - 3$ with $s = u + 1$. This is $t \leq u + \kappa - 3$. At $\kappa = 10$ it is the bound $t \leq u + 7$ in the proof of Theorem 12.1. Substituting t gives $3b + 2g + h + 3 \leq u$. Together with $u \geq 4 + 2\eta$ and $u \leq \kappa - 1$, a feasible u exists precisely when both lower bounds are at most $\kappa - 1$, which is (19).

If $\kappa = 10$, then $2(b + g) \leq 5$. Since $b \geq 2$, this forces $b = 2$ and $g = 0$. The second inequality becomes $6 + h \leq 6$, so $h = 0$. Then u runs from $\max(8, 9)$ to 9, hence $u = 9$ and $v = 14$.

If $\kappa = 11$, then $2(b+g) \leq 6$. The pair $(b, g) = (2, 0)$ allows $h \leq 1$. The pair $(3, 0)$ fails $9+h \leq 7$, and $(2, 1)$ fails $8 + h \leq 7$. Thus the solutions are

$$(b, g, h, u) \in \{(2, 0, 0, 9), (2, 0, 0, 10), (2, 0, 1, 10)\},$$

all with $v = 15$. The first is the excess-10 solution with a larger blue unit count. The other two are new.

If $\kappa = 12$, then $2(b + g) \leq 7$ and $3b + 2g + h \leq 8$. The tuple $b = 2$, $g = 1$, $h = 0$ satisfies both, and u runs from $\max(10, 11)$ to 11, so $u = 11$ and $v = 17$. The blue forest is $2K_{1,4} \sqcup K_{1,3} \sqcup 17K_2$. Lemma 12.2 requires every blue component of size at least 3 to have the same size a. This list has both 4 and 3.

The same system at $\kappa = 10$ had forced $h = g = 0$ and a single pair. That forcing used $u \leq 9$ and $\eta \leq 2$. Both fail as soon as $\kappa \geq 11$, since $\eta \leq \lfloor (\kappa - 5)/2 \rfloor$ and the right side is at least 3. □

Proposition 13.2. *Let $F_1 = F_2 = 20K_2$. Then $L = M = 39$ and $c = 20$. Let G be $K_{3,3}$, or a 5-cycle. In both cases $e(G) < L$. After disjoint edges are added so that the graph has $L - 1$ edges, the matching deficit is 7 for $K_{3,3}$ and 4 for the 5-cycle. Neither graph arrows (F_1, F_2). A disjoint union of cycles arrows no pair in which both heads are at least 3.*

Proof. Every diagonal equals 1, there are 39 of them, and $2 \cdot 20 - 20 = 20$.

The graph $K_{3,3}$ has 9 edges and matching number 3, so $\Phi(K_{3,3}) = 6$. Adding a disjoint edge raises e and ν by one and preserves Φ. Reaching $L - 1 = 38$ edges adds 29 edges and raises the matching number to 32. The deficit against $M = 39$ is 7. Equivalently, the padded deficit equals $\Phi - (L - M) + 1 = 7$.

A 5-cycle has 5 edges and matching number 2, so $\Phi = 3$. Reaching 38 edges adds 33 edges and raises the matching number to 35. The deficit is 4.

Theorem 5.2 gives $\hat{R}(20K_2, 20K_2) = 39$. Both hosts have fewer than 39 edges before and after the count 38, so neither arrows the pair.

Suppose both heads are at least 3 and H is a disjoint union of cycles. Every vertex has degree 2. In any red-blue coloring each color class has maximum degree at most 2, so neither color contains a star of size 3. The coloring avoids both forests. No choice of the component sizes, and no padding by later disjoint edges, makes this host arrow the pair. □

These two statements limit a substitution of κ for 10. They do not color every pair of excess κ.

Proposition 13.3. *Let M be a maximum matching of a finite simple graph G, and let H be the graph obtained by deleting the edges of M and retaining the vertices. If H has an edge, then $\nu(H) \geq 1$. In particular, H cannot both have matching number 0 and contain a cycle or a copy of $K_{3,3}$.*

Proof. An edge is a matching of size 1. A cycle of length k has matching number $\lfloor k/2 \rfloor \geq 1$, and $\nu(K_{3,3}) = 3$. □

Deleting the edges of a maximum matching preserves the identity $e(H) = \Phi(G)$. It forces $\nu(H) = 0$ only when H has no edges. A cycle or a copy of $K_{3,3}$ in H has positive matching number, so those blocks cannot be offered as the remainder of a maximum matching. A coloring of such a block also does not determine the colors of the restored matching edges. Proposition 13.2 shows that deficit 4 and deficit 7 occur among graphs with $L - 1$ edges. Those examples do not arrow. Forcing deficit 2 for every arrowing graph with fewer than L edges is not a consequence of coloring those blocks.

Proposition 13.4. *Let $F = P_3 \sqcup 11K_2$ and $J = 2K_{1,4} \sqcup K_{1,3} \sqcup 17K_2$. The diagonal sum of (F, J) is 80. Delete one edge from the copy of $K_{1,3}$ and call the resulting forest J^*, so $J^* = 2K_{1,4} \sqcup P_3 \sqcup 17K_2$. The diagonal sum of (F, J^*) is 79. If G arrows (F, J) and e is an edge of G at a vertex of degree at least 3, then $G - e$ need not arrow (F, J^*). If a blue P_3 is already present and e is restored in blue at its center, the blue component is a copy of $K_{1,3}$.*

Proof. Here $s = 12$ and $t = 20$. The only change of lists is that one blue size 3 becomes a blue size 2, and the diagonal sum falls by 1. Arrowing quantifies over all colorings of all edges. A coloring of $G - e$ does not see e, so it may avoid both F and J^* even though every coloring of G has a red F or a blue J. In the other direction, a blue P_3 plus one blue edge at its center is a blue $K_{1,3}$. If the rest of a blue J^* is vertex-disjoint from that star, the restored edge supplies the copy of $K_{1,3}$ required by J. □

Proposition 13.5. *The quantity $(D - 3)g + 17 - 9D/2 - 2p$ does not contain the unit count u. The relation $u = N - g - p$ determines N from u, g, and p. It does not determine the matching numbers of two factor-critical components.*

Proof. The displayed expression depends only on D, g, and p. Under the hypotheses of Lemma 8.3, including $u \leq 9$, the substitution $u = N - g - p$ removes u from the formal difference of (7) and (8). For $u \geq 11$ the inequality $5p \leq 12$ is no longer available. Raising u raises N, and the rank $N - 6$ in (7) moves with N. Lemma 8.3 does not force each new unit to add $D - 2$ to the lower bound and $D - 3$ to the upper bound. □

14. Why unit deletion does not finish the induction

A pair of excess $\kappa \geq 11$ that is not settled by Theorem 12.1 still has to be colored directly. The numerical exits around that pair are easy to state and are not a coloring of the remaining pair. If the largest head is at least $2\kappa - 9$, the pair is outside the range in which every head is at most $2\kappa - 10$. If the padded deficit d satisfies $d \leq 1$, or $d \geq 6$ and $\kappa \leq 2d + 7$, the small-deficit and large-deficit estimates used for excess at most 10 are the relevant statements, and they are not the assertion that every 2-coloring misses both forests merely because one embedding of a

nonunit core leaves too few edges. If $E_2 \geq 2t - c_0$ for some $c_0 < 4d$ and $t - s \geq \max(1, c_0 - 2d + 2)$, an arrowing host cannot have deficit d. A pair that meets none of these exits has both heads at most $2\kappa - 10$ and a bounded deficit. Deleting a unit from a target does not color it.

Proposition 14.1. *Let F_1' be F_1 with one component K_2 removed, and let G^* be G with one edge e removed. A red-blue coloring of G^* with no red F_1' may, after e is restored in red, contain a red F_1.*

Proof. The edge e is chosen in the uncolored graph G. It is not chosen from a red copy, because no coloring has been fixed. Let a coloring of G^* avoid red F_1'. If both ends of e lie outside every red component of that coloring, the restored red edge is a red K_2 disjoint from that copy of F_1', and the union is a red F_1. If instead one end of e is the center of a red star of size $n_i - 1$, the restored edge raises that star to size n_i. The inductive hypothesis supplies some avoiding coloring of G^*. It does not determine which of these two attachments occurs. □

Theorem 1.1 is proved when one forest is a matching, when one forest is a disjoint union of copies of P_3, and when the net unit excess is at most 10. Substituting κ for 10, forcing deficit 2 by deleting a maximum matching, shortening a $K_{1,3}$ to a P_3, and deleting a unit before the coloring exists do not prove it for an arbitrary excess.

References

[1] S. A. Burr, P. Erdős, R. J. Faudree, C. C. Rousseau, and R. H. Schelp, Ramsey-minimal graphs for multiple copies, *Indagationes Mathematicae* **40** (1978), 187–195.